7.3 Special Products of Polynomials

Name _____ Period ____ Date _____

Algebra I 7.3 · Student sheet

I can…

- use the square of a binomial pattern.
- use the sum and difference pattern.
- use special product patterns to solve real-life problems and do mental math.

Vocabulary · fill in the missing word

Square of a binomial — $(a + b)^2 =$ _____ and $(a - b)^2 = a^2 - 2ab + b^2$

Sum and difference pattern — $(a + b)(a - b) =$ _____

Part 1 · Pattern hunt

Multiply. Then look for a shortcut in each group.

Square of a sum	Product	Square of a difference	Product
$(x + 1)^2$		$(x - 2)^2$	
$(x + 4)^2$		$(x - 5)^2$	

Sum and difference	Product
$(x + 3)(x - 3)$	
$(x + 6)(x - 6)$	
$(2x + 1)(2x - 1)$	

Shortcuts: $(a + b)^2 =$ _____ $(a - b)^2 =$ _____ $(a + b)(a - b) =$ _____

Part 2 · Notes and examples (fill in each note, then work the example)

Square of a binomial (sum): $(a + b)^2 =$ _____

The middle term: The middle term of a squared binomial is _____ times the product of the two terms.

Example 1

Find the product.

$(x + 7)^2$

Square of a binomial (difference): $(a - b)^2 =$ _____

Example 2

Find the product.

$(2x - 5y)^2$

Sum and difference pattern: $(a + b)(a - b) =$ _____

The middle terms: In the sum and difference pattern, the middle terms _____.

Example 3

Find the product.

$(3m + 4)(3m - 4)$

Example 4

Find the product.

$(p - 2q)(p + 2q)$

Example 5

Use special products to find each product.

a. 51^2 b. $29 \cdot 31$

Example 6

A square has side length $(4x - 3)$. Write a polynomial that represents its area.

Part 3 · Partner practice

Partner 1

Find the product.

$(5x - 2)^2$

Partner 2

Find the product.

$(6 + y)(6 - y)$